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TAHUN 2026**



**TEST PROJECT
MODUL C – WINDOWS ENVIRONMENT
IT NETWORK SYSTEMS ADMINISTRATION**

Introduction

You have been appointed as the system administrator for **itnsa.id** and **lab.itnsa.id**. In this organization, Microsoft technologies are primarily used to provide centralized management, file services, and web services. At the same time, Ansible is used to automate system configuration, deployment, and routine administrative tasks.

Your role is to ensure all services are properly configured, integrated, and operational within the given time constraints. Attention to detail and time management are critical to your success in this assignment.

Senior System Administrator had already configure IP address on all machine & gave you certificate for service located at **C:\Certificate**

Also your Senior already configured Primary Domain Controller for **itnsa.id** domain on **SRV.itnsa.id** and Routing features on **EDGE.itnsa.id**. So, your job is only focus on creating resources for **itnsa.id** and configure child domain **lab.itnsa.id** on **DC.lab.itnsa.id**

For Ansible tasks, you can use pre-installed **Visual Studio Code** and **Zealdocs** on **ansible-srv**.

All configuration on Windows Server and Client **MUST BE PERSISTS AFTER REBOOT!**

There is a **Management** interface on each VM, **DO NOT TURN IT OFF!**

Credential Information

Windows

Username: User / Administrator

Password: Skill39!

Linux

Username: root / sysop

Password: Skill39!

Description of project and tasks

Basic Configuration

- 1. Configure **hostname** on all hosts refer to the information table
- 2. Install every certificate inside the **C:\Certificate** folder into **Local Machine**
 - For Root CA, install it on **Trusted Root Certification Authority**
 - For password protected certificates, use password **Skill39!**

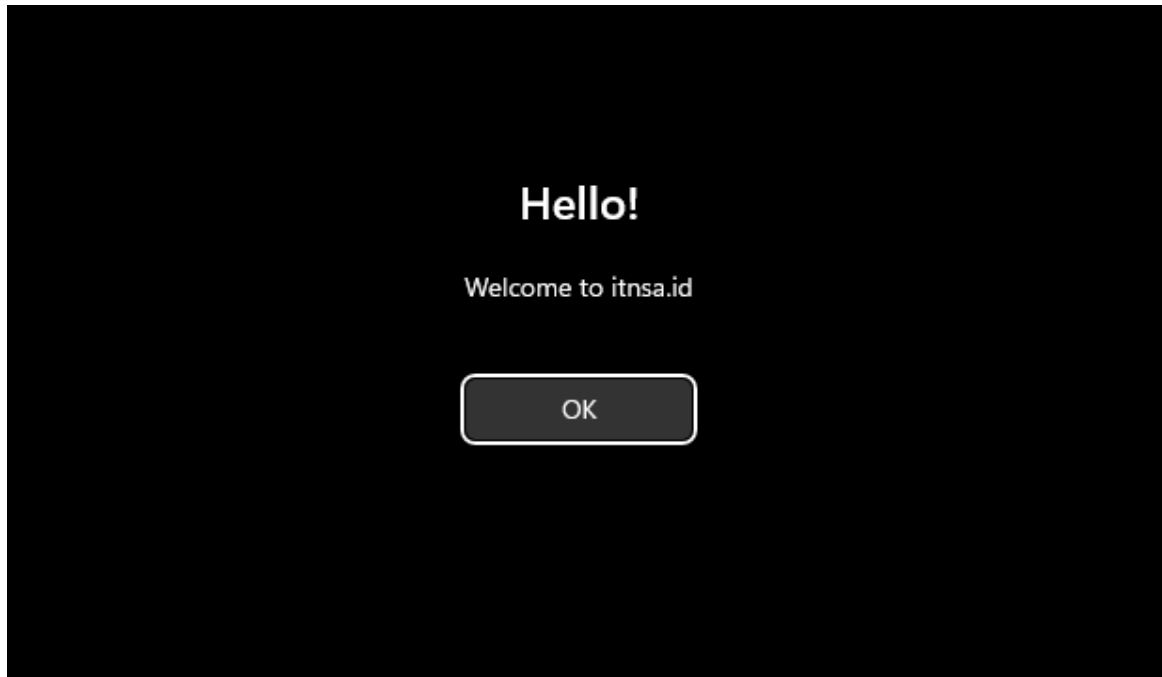
Active Directory

SRV.itnsa.id

- 1. Configure Active Directory Sites and Services:
 - a) Rename default site into **ITNSA-ID**
 - b) Create and associate the following IP subnets with the **ITNSA-ID** site:
 - o **192.168.1.0/24**
 - o **192.168.2.0/24**
 - c) Only **SRV** server assigned to the **ITNSA-ID** site
- 2. Create the following Organizational Units (OU) inside the itnsa.id domain:
 - a) **Employee**
 - b) **Finance**
 - c) **Engineer**
 - d) **Manager**
- 3. Create AD Group with specification refer to the list below:
 - a) **Employee** group created inside **Employee** OU
 - b) **Finance** group created inside **Finance** OU
 - c) **Engineer** group created inside **Engineer** OU
 - d) **Manager** group created inside **Manager** OU
- 4. Create Active Directory user accounts according to the table below. Use **PowerShell script** to create all user accounts. Use password **Skill39!** and make sure users **do not need to change their password at the next logon**.

Username Format	Total Accounts	Target OU
employee-1 ~ employee-100	100	Employee
engineer-1 ~ engineer-58	58	Engineer
finance-1 ~ finance-21	21	Finance
manager-1 ~ manager-10	10	Manager

6. Configure group membership for all created users. Each user must be a member of the security group that corresponds with their OU (e.g., all users in the **Finance** OU must be members of the **Finance** group).
7. Configure **SRV.itnsa.id**, **EDGE.itnsa.id** and **WORKSTATION.itnsa.id** as the domain members of **itnsa.id**
8. Configure group policy called **Banner** with specification below:
 - Set the login banner title into **Hello!**
 - Set the login banner text into **Welcome to itnsa.id**



9. Configure group policy called **No-Animation** with specification below:
 - Disable the first sign-in animation for users logging onto the workstation.

DC.lab.itnsa.id

1. Promote this server as a new domain controller for a new child domain named **lab.itnsa.id** in the existing **itnsa.id** forest.
2. Configure Active Directory Sites and Services:
 - a) Create a new site named **LAB-ITNSA-ID**
 - b) Create and associate the IP subnet **10.1.1.0/24** with the **LAB-ITNSA-ID** site.
 - c) Only **DC** server assigned to the **LAB-ITNSA-ID** site.
3. Inside the **lab.itnsa.id** domain, create the following Organizational Units (OU):
 - a) **Operator**
 - b) **Member**
4. Inside each created OU, create a security group with the same name as the corresponding OU.

5. Create Active Directory user accounts according to the table below. Use password **Skill39!** and make sure users **do not need to change their password at the next logon**.

Username Format	Total Accounts	Target OU
operator1 - operator2	2	Operator
member1 - member10	10	Member

6. Configure group membership for all created users. Each user must be a member of the security group corresponding to their OU.

DNS Service

SRV.itnsa.id

1. Create the following Forward DNS records in the **itnsa.id** zone:

Type	Record Name	Value / Target
A	DC	192.168.1.1
A	SERVER-GW	192.168.1.254
A	EDGE	192.168.1.254
A	CLIENT-GW	192.168.2.254
CNAME	csdrive	EDGE.itnsa.id
CNAME	file	EDGE.itnsa.id
CNAME	www	SRV.itnsa.id
CNAME	internal	SRV.itnsa.id
CNAME	extra	SRV.itnsa.id

2. Create the following Static Reverse DNS (PTR) records:

Record Name	IP Address
SRV.itnsa.id	192.168.1.1
EDGE.itnsa.id	192.168.1.254
EDGE.itnsa.id	192.168.2.254
CLIENT-GW.itnsa.id	192.168.2.254

3. Make sure **WORKSTATION** IP address automatically created on reverse DNS zone.

DC.lab.itnsa.id

1. Create the following Forward DNS records in the **lab.itnsa.id** zone:

Type	Record Name	Value / Target
A	DC	10.1.1.10
A	EDGE	10.1.1.254

2. Configure a DNS forwarders pointing to the IP address of **SRV.itnsa.id**.

DHCP Service

EDGE.itnsa.id

1. Install and configure DHCP services on this server. Create a new DHCP scope with the following specifications:

Scope Name	itnsa.id_Client
Range	192.168.2.10 - 192.168.2.200
Subnet Mask	255.255.255.0
Excluded Address Range	192.168.2.110 - 192.168.2.120
DNS Servers	192.168.1.1, 10.1.1.10
Domain Name	itnsa.id
Gateway	192.168.2.254
Lease Time	1 day, 1 hours, 39 minutes

2. Configure the DHCP server to always perform dynamic DNS updates for clients.
3. Authorize this server in Active Directory as an authoritative DHCP server for the **itnsa.id** domain.

IIS Web Service

SRV.itnsa.id

1. Install and configure the **Internet Information Services (IIS)** web server role on this server.
2. All websites created on this server must listen on **HTTPS** port and use **itnsa.id** certificate located on **C:\Certificate** directory

Note:
*Certificate is wildcard certificate, so it can be used to deploy any website with *.itnsa.id domain!*

3. Use the provided website content files located in the **C:\Content** directory for each website on the table.
4. Create and configure websites refer to the specifications in the table below:

Website Name (Binding)	Path	Default Index File
internal.itnsa.id	C:\www\internal	Internal.html
www.itnsa.id	C:\www\public	Index.html
extra.itnsa.id	C:\www\extra	Extra.html

5. Configure Windows Authentication for **internal.itnsa.id** and make sure every domain users can access this site after authentication.

DC.lab.itnsa.id

1. Install **Internet Information Services (IIS)** web server role on this server for future website deployment using playbook. You can set anything for website content.
2. For easier management, install and configure IIS Management Service.

Shared Folder

EDGE.itnsa.id

1. Create a new root directory for all shared folders at the following path: **C:\Shared Folder**
2. Create two shared folders refer to the list below:
 - Shared folder named **Home** pointing to the **C:\Shared Folder\Home** directory. This share will be used for future deployment..
 - Shared folder named **Group** pointing to the **C:\Shared Folder\Group** directory. This share will be used for group-specific data.

3. On all created shared folders above, remove the default **Everyone** account share permission and set **Domain Users** to **Full**
4. On **Group** shared folder, create folder with NTFS permission refer to the list below:
 - **Employee** directory with **Employee & Manager** group has Full permission. Other groups must be prohibited.
 - **Engineer** directory with **Engineer & Manager** group has Full permission. Other groups must be prohibited.
 - **Finance** directory with **Finance & Manager** group has Full permission. Other groups must be prohibited.
 - **Manager** directory with **Manager** group has Full permission. Other groups must be prohibited.

File Server Resource Manager

EDGE.itnsa.id

1. Configure Quota Management to enforce storage limits:
 - Apply a **100MB** hard limit quota directly to **each group folder** inside the **Group** shared folder.
2. Configure File Screening to control file types:
 - On the **Home** folder, apply a file screen to **block executable** files from being saved.
 - On the **Group** folder, apply a file screen that **only permits** files with the **.txt** and **.doc** extensions to be saved.

Distributed File System

EDGE.itnsa.id

1. Configure a new domain-based DFS Namespace with name **CSDrive**
2. Create the following folders within the **CSDrive** namespace with their respective targets:
 - **\\itnsa.id\CSDrive\Home** targeting **\\EDGE.itnsa.id\Home**
 - **\\itnsa.id\CSDrive\Group** targeting **\\EDGE.itnsa.id\Group**

Ansible Automation

ansible-srv

The **ansible-srv** host is pre-configured with the necessary inventory and credentials to connect to the Windows servers in the **itnsa.id** domain. You can check it in the **/opt/ansible** directory. **DO NOT MODIFY THE INVENTORY FILE!**

Your task is to create and execute Ansible playbooks to automate the following administrative tasks. All playbooks should be created in the **/opt/ansible** directory with **.yml** format.

1. Intro Playbook

As an introduction, create a playbook named **/opt/ansible/intro.yml**. This playbook must perform the following actions on all Windows hosts defined in the inventory file:

- Ensure the directory **C:\ITNSA** exists
- Create a file at **C:\ITNSA\ansible_ready.txt**
- Install **Telnet-Client** feature

2. Shared Folder Deployment

Create a playbook named **/opt/ansible/deploy_project.yml** that targets **EDGE.itnsa.id**.

This playbook must do following tasks:

- Create **C:\Project** directory as the root directory for shared project
- Inside project directory create **Project1**, **Project2** and **Project10**
 - Create SMB share with name same as the directory name
 - Grant **Domain Users** Full access permission

3. Website Deployment

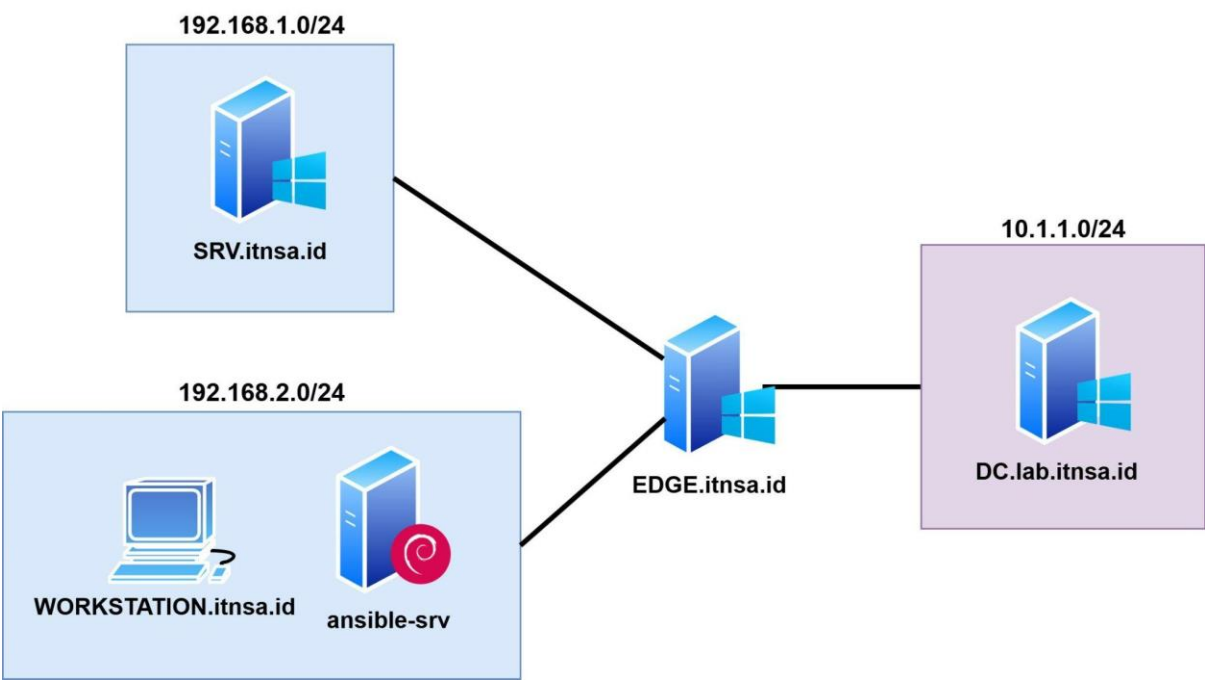
Create a playbook named **/opt/ansible/deploy_website.yml** that targets **SRV.itnsa.id**.

This playbook must be able to deploy a new website on **SRV.itnsa.id**.

- The created playbook must perform this step:
 1. Create a **DNS A record** in the **itnsa.id** zone called **ansible-web** that resolve to IP address of **SRV.itnsa.id**
 2. Create a root directory for the website at **C:\project_website\ansible_web**
 3. Configure default index content to **"This is Ansible website"**
 4. Create a new IIS website that listens on HTTP and must be accessible from the client.

Appendix

Topology



Addressing Table

Hostname	FQDN	IP Address
SRV	SRV.itnsa.id	Ethernet0: 192.168.1.1/24 Management: 172.20.0.171/24
EDGE	FW.itnsa.id	Ethernet0: 192.168.1.254/24 Ethernet1: 192.168.2.254/24 Ethernet2: 10.1.1.254/24 Management: 172.20.0.172/24
DC	DC.lab.itnsa.id	Ethernet0: 10.1.1.10/24 Management: 172.20.0.173/24
WORKSTATION	WORKSTATION.itnsa.id	Ethernet0: DHCP Management: 172.20.0.174/24
ansible-srv	N/A	ens192: 192.168.2.248/24 ens256: 172.20.0.175/24